Question Paper

Exam Date & Time: 01-Jan-2025 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.Tech. (ELECTRONICS & COMMUNICATION ENGINEERING) EXAMINATIONS - DECEMBER 2024 / JANUARY 2025
SUBJECT: ECE 3122/ECE_3122 - MICROPROCESSORS
(MAKEUP)

Marks: 50 **Duration: 180 mins.**

Answer all the questions.

1A)	Describe the function of each bit in the CPSR of ARM 7 with neat bit diagram. Briefly discuss the difference between the current PSR and the saved PSR.	(4)
1B)	Elaborate with a diagram the banked registers available for ARM7 processor in all its modes.	(3)
1C)	Describe the function of 'macro' directive with proper syntax and example. Explain the advantages and disadvantages of using it in the ARM7 coding.	(3)
2A)	Discuss the function of following ARM7 instructions in detail with proper syntax and relevant example. i) BLX ii) SMALL iii) TST iv) MSR	(4)
2B)	a) What is the difference between the followings? i) Full stack and Empty stack ii) Ascending stack and Descending stack b) Assuming the same PRE conditions for both the instructions, trace the associated memory and registers content POST execution for the following ARM7 instructions: SP=0xE0000100 R0=0x12345678 R2=0x19283746 R4=0xABCDEF98 i) STMFD SP!, {R0, R2, R4} ii) STMEA SP!, {R0, R2, R4}	(3)

2C)	Write an embedded C program instructions equivalent to the ARM7 code given below:
	FUNC MOV R5, #2
	MOV R4, #5
	SUB R0, R4, R5
	MOV PC, LR
	MAIN MOV R3, #1
	BL FUNC
	ADD R2, R0, R3
		(3)

3A)	Write an assembly language program in ARM 7 to find the sum of digits in given 8-digit hexadecimal number. Assume that the number is stored in the memory. Store the result in the memory location. Explain the logic employed in computing.	(4)
3B)	Determine the 32-bit instruction code for the following ARM7 instruction. Describe the significance	(3)

of all the fields in the operation code.
ORRCS R0, R9, R6, LSR R11

3C)	Compare and contrast between subroutines and macros in assembly language programming. Explain with example.	(3)
4A)	A LED and switch is interfaced to LPC2148 through the port pins P0.0 and P0.16 respectively. Write an embedded C program with comments to explain the logic for turning LED ON or OFF depending on the status of the switch. Draw the interface diagram.	(4)
4B)	Write the features of fast GPIO registers. Give the functionality of the following registers: FIOxDIR, FIOxMASK, FIOxPIN, FIOxSET	(3)
4C)	Discuss the features of LPC2148 DAC and explain how this can be programmed to do conversion of digital data into analog output.	(3)
5A)	Discuss significance of vector table and vector address in ARM 7. Describe the commonly used instructions that makes a branch to the interrupt handler.	(4)
5B)	With neat block diagram, describe the branch instruction datapath activity in ARM 7 processor.	(3)
5C)	What is the significance of bit band memory in Cortex-M3 core? Calculate the bit-band alias address of 3^{th} bit of the bit-band byte present at the address 0x20012345, in Cortex M3.	(3)

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